# Limits

Questions 1–29

**2.09 — Question 1.** Find

a) $\lim_{x\to\infty}(1+e^{-x})$

b) $\lim_{x\to 0}(x^4e^x)$

**2.09 — Question 2.** Evaluate each of the following:

a) $\lim_{x\to 0}\sqrt{\dfrac{x^3-2x^2+10}{4x^3+1}}$

b) $\lim_{x\to 0}\dfrac{x+4}{\sqrt{3x^2-5x+2}}$

c) $\lim_{x\to 0}\dfrac{\sqrt{2x^3-5x^2-1}}{x^2-2}$

**2.09 — Question 3.** Find each of the following:

a) $\lim_{n\to\infty} \dfrac{5-4n}{n}$

b) $\lim_{n\to\infty} \dfrac{2n-3}{n^2}$

c) $\lim_{n\to\infty} \dfrac{n+7}{2n+1}$

d) $\lim_{n\to\infty} \dfrac{10-3n}{n-3}$

**2.09 — Question 4.** Evaluate these limits:

a) $\lim_{n\to\infty} \dfrac{2n^2-4n+5}{3n^2+n-4}$

b) $\lim_{n\to\infty} \dfrac{5n^2+2n+1}{2n^2-n-7}$

c) $\lim_{n\to\infty} \dfrac{5n^2-4n+5}{4-n-7n^2}$

**2.09 — Question 5.** Find each of the following:

a) $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$

b) $\lim_{x \to 1} \dfrac{x^3 - 1}{x - 1}$

**2.09 — Question 6.** Find each of the following:

a) $\lim_{x \to 3} \dfrac{\sqrt{x} - \sqrt{3}}{x - 3}$

b) $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$

c) $\lim_{x \to k} \dfrac{\sqrt{k} - \sqrt{x}}{x - k}$, for positive constant $k$

**2.09 — Question 7.** Use small angle approximations to show that

a) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right) = 1$

b) $\lim_{x\to 0}\left(\frac{\cos x - 1}{x}\right) = 0$

**2.09 — Question 8.** Find each of the following:

a) $\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{1-\cos 4x}{x^2}\right)$

c) $\lim_{x\to 0}\left(\frac{4\cos x}{\sin x + \cos x}\right)$

**2.09 — Question 9.** By using the approximation $\ln(1+\frac{x}{2}) = \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{24}$ where $x$ is small, find $\lim_{x\to 0}\left(\frac{8\ln(1+\frac{x}{2})}{x}\right)$.

**2.09 — Question 10.** By using the approximations $\cos 2x \approx 1 - 2x^2$ and $\sin 3x \approx 3x - \frac{9}{2}x^3$ for small values of $x$, find $\lim_{x\to 0}\left(\frac{3x\cos 2x - \sin 3x}{3x^3}\right)$.

**2.09 — Question 11.** By considering the first 4 terms in the expansion of $\sqrt{4+x}$, find $\lim_{x\to 0}\left(\dfrac{-2+\sqrt{x+4}}{x}\right)$.

**2.10 — Question 1.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 2}\left(\dfrac{x^2-4}{x-2}\right)$

b) $\lim_{x\to 1}\left(\dfrac{\ln x}{x-1}\right)$

c) $\lim_{x\to 0}\left(\dfrac{e^x-1}{x}\right)$

d) $\lim_{x\to 0}\left(\dfrac{e^{2x}-1}{x}\right)$

e) $\lim_{x\to 0}\left(\dfrac{e^x-1}{\ln(1+x)}\right)$

f) $\lim_{x\to 0}\left(\dfrac{\ln(1+3x)}{x}\right)$

**2.10 — Question 2.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{1-\cos x}{x}\right)$

c) $\lim_{x\to 0}\left(\frac{x}{\sin x}\right)$

d) $\lim_{x\to 0}\left(\frac{x^2}{\sin x}\right)$

e) $\lim_{x\to 0}\left(\frac{-x+\tan x}{x}\right)$

f) $\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)$

**2.10 — Question 3.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)$

b) $\lim_{x\to 0}\left(\frac{x-\sin x}{x^3}\right)$

c) $\lim_{x\to \infty}\left(\frac{\ln x}{x}\right)$

d) $\lim_{x\to \infty}\left(\frac{x^2}{e^x}\right)$

e) $\lim_{x\to 0}\left(\frac{e^{3x}-1-x}{x^2}\right)$

f) $\lim_{x\to 0}\left(\frac{\ln(1+x)-x}{x^2}\right)$

g) $\lim_{x\to \infty}\left(\frac{\ln x}{\ln(x+1)}\right)$

**2.10 — Question 4.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{\sinh x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{\tanh x}{2x}\right)$

c) $\lim_{x\to 0}\left(\frac{\sinh x - x}{4x^3}\right)$

**2.10 — Question 5.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{\arcsin x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{\arctan x}{x}\right)$

**2.10 — Question 6.** A student tried to find the value of $\dfrac{\cos x + 1}{e^x + 2}$ as $x \to 0$. They wrote:

$$\lim_{x\to 0}\left(\frac{\cos x + 1}{e^x + 2}\right) = \lim_{x\to 0}\left(\frac{-\sin x}{e^x}\right) = 0.$$

a) Explain the student's mistake.
b) Find the correct value of the limit.

**2.10 — Question 7.** Use L'Hôpital's rule to evaluate $\lim_{x\to\infty}(xe^{-2x})$.

**2.10 — Question 8.**

a) Rewrite $x\ln(1+\frac{1}{x})$ as a quotient for the use of L'Hôpital's rule.

b) Hence, use L'Hôpital's rule to find $\lim_{x\to\infty}\left(5x\ln(1+\frac{1}{x})\right)$.

**2.10 — Question 9.** Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{xe^x}{\sin x}\right)$

b) $\lim_{x\to 0}\left(\frac{e^x\sin 7x}{\ln(4x+1)}\right)$

c) $\lim_{x\to 0}\left(\frac{\sin x - x\cos x}{x^3}\right)$

d) $\lim_{x\to 0}\left(\frac{1-\cos 5x+3x\sin 3x}{2x^2}\right)$

**2.10 — Question 10.** Use L'Hôpital's rule to find:

a) $\lim_{x\to\pi}\left(\dfrac{\sin x}{x-\pi}\right)$

b) $\lim_{x\to\pi}\left(\dfrac{x\cos x+\pi}{(x-\pi)^2}\right)$

c) $\lim_{x\to\pi}\left(\dfrac{\cos(\frac{x}{2})}{x\sin 2x}\right)$

**2.10 — Question 11.** Use L'Hôpital's rule to find:

a) $\lim_{a\to\infty}\left(\dfrac{a}{e^a}\right)$

b) Hence find $\displaystyle\int_0^\infty xe^{-x}\,dx$.

**2.10 — Question 12.** Use L'Hôpital's rule to find:

a) $\lim_{a\to\infty}\left(\frac{\ln a}{a}\right)$

b) Hence find $\int_1^{\infty}\frac{\ln x}{x^2}\,dx$.

**2.10 — Question 13.**

a) Show that $\frac{d}{dx}[\ln(\cos x)] = -\tan x$.

b) Hence use L'Hôpital's rule to evaluate $\lim_{x\to 0}\left(\frac{\ln(\cos x)}{x^2}\right)$.

**2.10 — Question 14.**

a) Show that $\dfrac{d}{dx}[\ln(\cosh x)] = \tanh x$.

b) Hence use L'Hôpital's rule to evaluate $\lim\limits_{x\to 0}\left(\dfrac{\ln(\cosh x) - 7x^2}{x^2}\right)$.

**2.10 — Question 15.**

a) Use L'Hôpital's rule to show that $\lim\limits_{x\to 0^+}(x\ln x) = 0$.

b) Explain why we must consider $x \to 0^+$ and not $x \to 0^-$.

**2.10 — Question 16.** The displacement $s$ of a particle is given by $s(t) = \dfrac{e^t - 1}{t}$. Use L'Hôpital's rule to find the velocity of the particle at $t = 0$.

**2.10 — Question 17.** Given that, for $x \geq 0$, $f(x) = x^{-3}(\ln(1+x) - x + \dfrac{1}{2}x^2)$:

a) Use a Maclaurin expansion to show that $\ln(1+x) - x + \dfrac{1}{2}x^2 \approx \dfrac{1}{3}x^3$.

b) Hence deduce that $\lim\limits_{x \to 0}(x^{-3}[\ln(1+x) - x + \dfrac{1}{2}x^2]) = \dfrac{1}{3}$.

c) Use L'Hôpital's rule to confirm the result in part (b).

d) Explain why L'Hôpital's rule alone is not sufficient to determine the behaviour of $f(x)$ near $x = 0$.

**2.10 — Question 18.** By considering where $y = (1 + \frac{1}{x})^x$, use L'Hôpital's rule to find $\lim_{x \to \infty} (1 + \frac{1}{x})^x$.